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Matomonized UD Japanese: A Revised Treebank Conforming to the UD Guidelines

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1 Introduction

The aim of the present paper is to address the fundamental issues pertaining to Universal Dependencies (UD) and its application to Japanese (UD Japanese; henceforth UDJ), and to propose a new annotation framework in which the major drawbacks of the existing UDJ are resolved by means of a Python-based pipeline that automatically converts the existing annotation to a linguistically more plausible and cross-linguistically coherent one. This pipeline, which takes an existing UDJ corpus as input and produces a revised corpus as output, will be referred to as the “Matomonizer”¹, and the resulting converted corpus will be called Matomonized UDJ (MUDJ). The present paper specifically aims to introduce the Matomonizer and discusses the key design decisions and implementational challenges involved in the first phase of this project: the conversion of the existing UDJ to MUDJ for Standard Japanese (SJ).

2 UD

Universal Dependencies (UD) is a framework for cross-linguistically consistent annotation of morphology and syntax based on dependency grammar. Its core principles include:

- (1) a. prioritizing cross-linguistic parallelism so that the same construction receives the same analysis across languages
- b. adopting a lexicalist approach in which syntactic relations hold between words, not morphemes
- c. favoring content words as heads, treating function words as dependents
- d. defining annotation units in terms of syntactic words rather than orthographic or phonological words.

A corpus of a language that follows the UD framework and annotation schema and that has been officially accepted by the UD project is called a **treebank**. As of UD v2.17, the project comprises over 300 treebanks for more than 180 languages. Japanese also participates in the UD project by depositing its own UD treebanks (see Section ?? below).

Typical and successful cross-linguistic studies using UD treebanks include research on *linguistic complexity* (Berdicevskis et al. 2018; Bentz et al. 2023), which tests the famous but impressionistic hypothesis known as the Complexity Trade-off Hypothesis (Hockett 1958, among others). This hypothesis predicts that there is a complexity trade-off between morphology and

¹*Matomo=na* (/matomo=na/) is a Japanese adnominal adjective meaning ‘proper’ or ‘decent’.

syntax such that languages with a complex morphological system generally have less complex syntactic systems, and vice versa.

A UD-based cross-linguistic study such as the one above is made possible only if each corpus is **standardized**, i.e., annotated in accordance with the UD general policy and guidelines. UD thus requires each treebank to comply with its standardizing policy and guidelines, as will be discussed in the following sections.

3 Basics of UD annotation

There are three core concepts in UD standardized annotation, all of which center on the fundamental principle of UD that **annotation should be strictly word-based**.

- (2) a. Word-internal morphological annotation is done via FEATS, which specifies what grammatical categories are encoded in the word.
- b. The part of speech of a word is indexed with the universal set of POS tags called UPOS.
- c. The dependency relation between words is specified with the universal set of dependency tags called DepRel.

As an initial approximation, Figure 1 illustrates how the English sentence *She left him a note* is annotated using these three annotation tags.

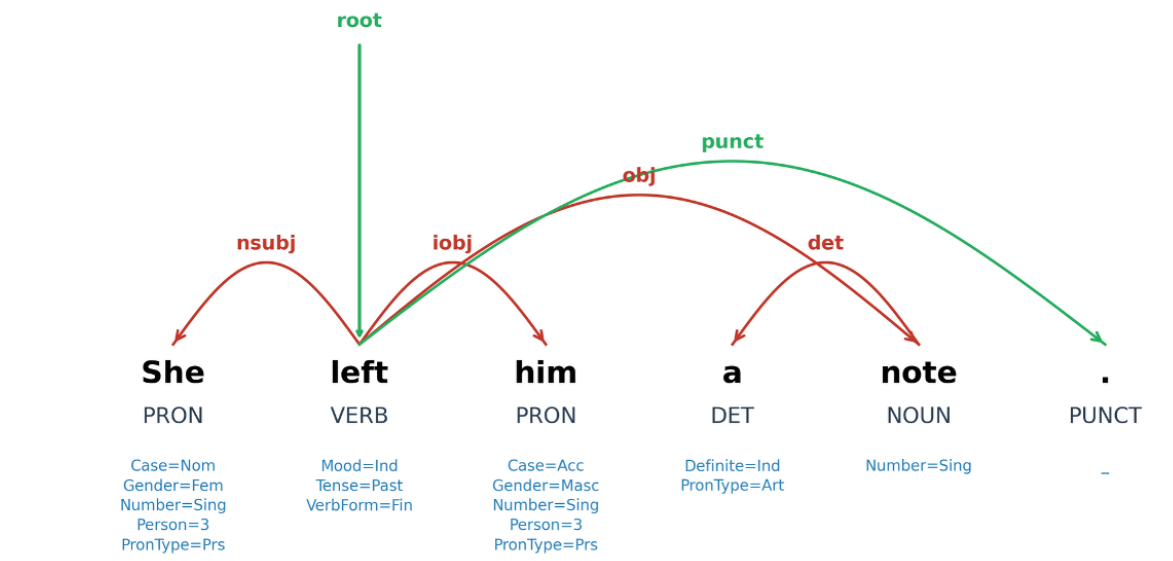


Figure 1: English *She left him a note* and its UD annotation: UPOS is indicated with capital letters. DepRel is indicated by a red arrow from the head to its dependent with the type of relation indicated above the arrow. FEATS is indicated under each word in blue text.

4 Fundamental Problems in UD Japanese

There are several Japanese UD treebanks (UDJ) as of 2026 (Table 1), drawn from different corpora and with slightly different annotation schemata.

UDJ treebanks are based on either SUW (Short-Unit Word) tokenization or LUW (Long-Unit Word) tokenization, as illustrated in Figure 2. Neither tokenization approach, however,

Table 1: A family of UDJ treebanks (as of 2026)

SUW-based treebanks

Treebank	Sentences	Tokens	Source Corpus
BCCWJ	57,109	1,253,903	Balanced Corpus of Contemporary Written Japanese
GSD	8,100	193,654	Google UDT 2.0 (Wikipedia)
KTC	7,158	189,943	Kyoto Text Corpus (Mainichi Newspaper)
PUD	1,000	28,788	Parallel UD (CoNLL 2017 shared task)
Modern	822	~14,000	Corpus of Historical Japanese (Meiji-Taishō)

LUW-based treebanks

Treebank	Sentences	Tokens	Source Corpus
BCCWJLUW	57,109	995,632	Same as BCCWJ, LUW segmentation
GSDLUW	8,100	150,243	Same as GSD, LUW segmentation
PUDLUW	1,000	22,910	Same as PUD, LUW segmentation

conforms to the word-based tokenization principle prescribed by UD: the “word” in UDJ’s terms is based on *Gakkō Bunpō* (GB; school grammar), a *kana*-based grammatical framework. Figure 2 provides an initial confirmation that the word segmentation of UDJ (whether SUW or LUW) is wildly divergent from what would be expected for word segmentation in general linguistics.

We identify three fundamental problems in the current UDJ treebanks, which we illustrate throughout this section using a single example: the adjective predicate *takaku nakatta* (‘was not expensive’). This simple expression is sufficient to reveal all three issues simultaneously.

4.1 Tokenization

The most fundamental problem with UDJ is that it tokenizes at the morpheme level rather than the word level. The UD guidelines explicitly state that “dependency relations hold between words” under a lexicalist approach, and that “morphological features are encoded as properties of words and there is no attempt at segmenting words into morphemes” (Marneffe et al., 2021). Nevertheless, UDJ adopts the Short Unit Word (SUW) as its token unit, splitting inflectional morphemes into separate tokens.

As shown in Figure 3, *takaku nakatta* is tokenized into three morpheme-level tokens in the current UD Japanese GSD: *taka-ku* (ADJ), *nakatt* (AUX), and *ta* (AUX). The negative morpheme and the past tense morpheme are each treated as independent syntactic tokens, both attached to the adjective via the aux relation. Under word-based tokenization consistent with the UD guidelines, *nakatta* should instead be treated as a single AUX token, yielding a two-token analysis: *taka-ku* (ADJ) and *na-kaQta* (AUX). This is directly parallel to how English treats *wasn’t* as a single token in UD English treebanks.

One additional comment on the “morpheme-level” tokenization of UDJ is that UDJ is not truly morpheme-based either. ☒ is treated as a single token, but it is not a morpheme: it consists of two morphemes, /taka/ and /ku/. UDJ is **lexeme-based**, in the sense that a lexeme (e.g., ☒) and its word-forms (e.g., ☒) are treated as if they were single morphemes. In linguistics, a lexeme is an abstract unit whose concrete realizations are the target of morphological analysis, but in GB, what is called morphological analysis is targeted at each lexeme (Shimoji 2020).

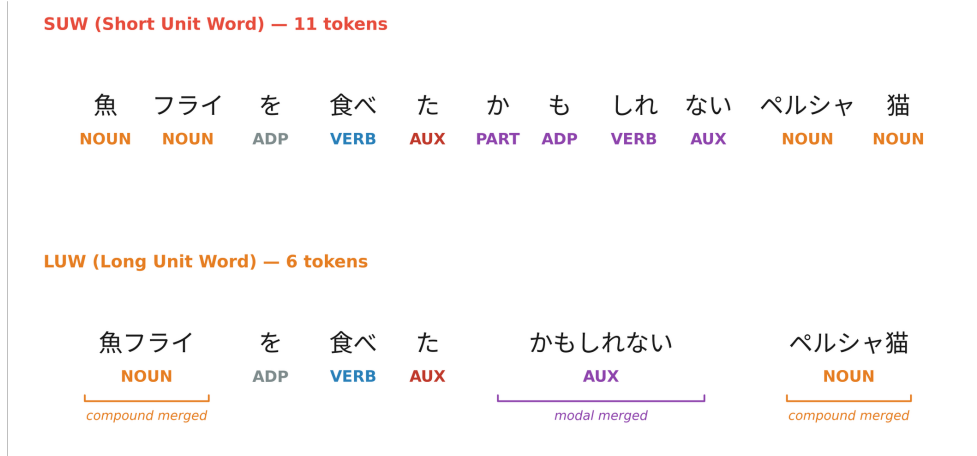


Figure 2: Comparison of SUW and LUW tokenization for the sentence *sakana hurai o tabeta kamo sirenai perusya neko* (‘a Persian cat that might have eaten fried fish’; from [Omura and Asahara 2018](#)). LUW merges compound nouns and modal expressions but leaves verbal inflection (e.g., *tabe* + *ta*) split at the morpheme level, with no morphological features annotated in either scheme.

Thus, the tokenization problem of UDJ stems not only from its poor understanding of what a word is but also from its unique interpretation of what a morpheme is. In UDJ, standardization is completely disregarded.

4.2 Feature Annotation

The “morpheme-level” (or lexeme-level, or whatever level lower than the word level) tokenization leads directly to a second problem: the near-complete absence of morphological feature annotation (with FEATS tags). Since FEATS concerns word-internal details, typically grammatical-feature information such as tense, aspect, mood, polarity, etc., there is no opportunity to use FEATS in UDJ, where almost all “words” correspond to morphemes and thus never have internal structure (Figure 4). The past tense 食 and the negative 食べない are analyzed as two “words” in UDJ and are annotated using syntactic annotation tags *UP0S* and *DepRel*. However, since they are clearly word-internal elements, the UD standard requires them to be regarded as grammatical features of the word /nakaQta/, as illustrated in the MUDJ version of Figure 4.

The difference in annotation between UDJ and MUDJ is not trivial. Rather, switching from UDJ to MUDJ has crucial consequences for cross-linguistic comparability regarding morphological complexity. UDJ lacks FEATS almost entirely, which gives linguists the false impression that Japanese is a near-isolating language—a fallacy rooted in the flawed annotation policy of UDJ. Had more precise annotation been applied to Japanese, as in MUDJ, this kind of problem could have been avoided. We will return to this issue in Section 5.

4.3 Dependency Relation Specification

The third problem concerns the poor specification of dependency relations (*DepRel*), which constitute the most informative part of UD, as is obvious from the name “Universal **Depend**-**encies**”. Consider the sentence *Hanako was not a student* and its Japanese equivalent *hanako=wa gakusee=de=wa na-kaQta*.

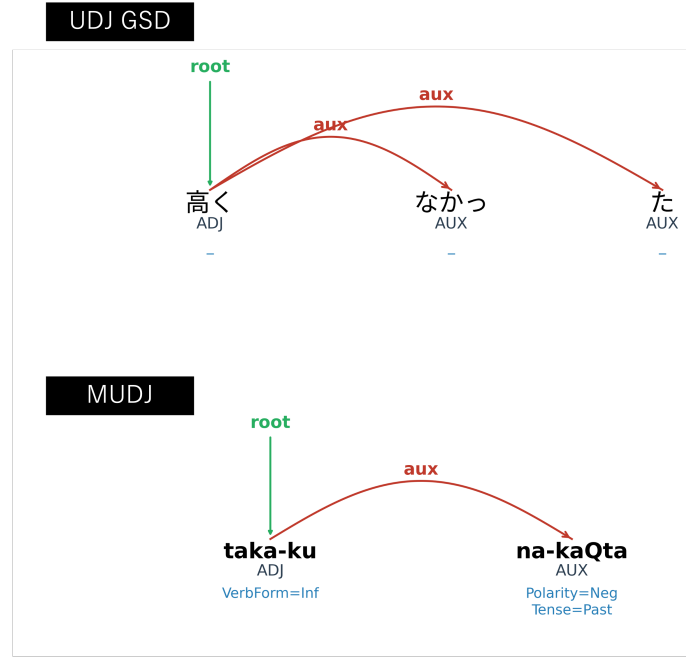


Figure 3: Tokenization problem. Top: the current UD Japanese GSD splits *takaku nakatta* into three morpheme-level tokens. Bottom: the proposed word-based tokenization produces two tokens, consistent with UD’s lexical-ist approach.

In UD English (upper panel of Figure 5), each function word receives a distinct label that reflects its grammatical role: the copula *was* is labeled *cop*, the negator *not* is labeled *advmod*, and the determiner *a* is labeled *det*. These four distinct *DepRel* labels—*nsubj*, *cop*, *advmod*, and *det*—immediately reveal the syntactic function of every word in the sentence.

In the current UD Japanese GSD (middle panel of Figure 5), however, the same sentence is annotated with only three types of *DepRel* for function words: *nsubj*, *case*, and *aux*. The copula *de*, the negative morpheme *nakatt*, and the tense morpheme *ta* all receive the identical label *aux*, despite serving entirely different grammatical functions.

This uniform labeling is not the result of careful linguistic judgment on the part of the annotators but simply reflects the fact that these bound forms are traditionally called 助詞 (*zyodoosi*) in Japanese school grammar, an unfortunate translation of which is “auxiliary”.

The consequence is that crucial functional distinctions maintained in other UD treebanks are collapsed, making it impossible to extract, for example, all copular predicates or all negated predicates through dependency labels alone. In MUDJ (lower panel of Figure 5), these distinctions are restored: the copula *=de* receives *cop*, the topic markers *=wa* receive *advmod* (with UPOS reassigned from *ADP* to *PART*), and the negative auxiliary *na-kaQta*—now a single word-level token with *Polarity=Neg|Tense=Past*—receives *aux*. The result is a four-way *DepRel* distinction (*nsubj*, *cop*, *advmod*, *aux*) that directly parallels the English annotation, enabling meaningful cross-linguistic comparison.

While UDJ seems indifferent to *DepRel*, it often makes cross-linguistically anomalous use of *DepRel* annotations, making cross-linguistic comparability even harder to achieve. For example, the topic marker *wa* in Figure 5 is labeled *case*, the same label used for genuine case particles such as the nominative *ga* and the accusative *o*.

As Asahara et al. (2018) themselves acknowledge, the current UDJ annotation assigns dependency relations based solely on surface forms without attempting to capture deeper

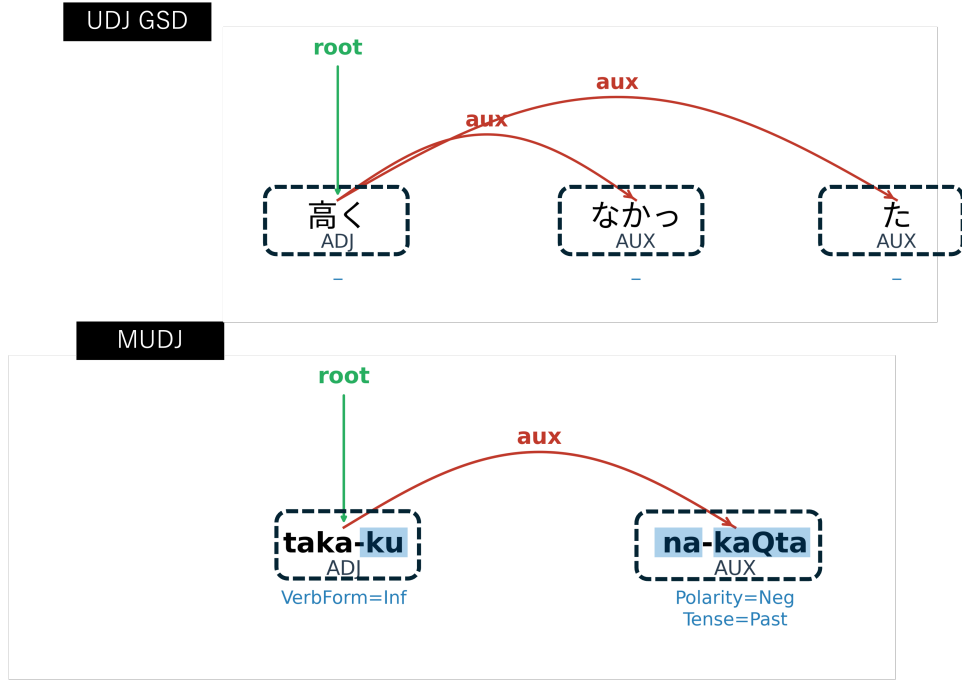


Figure 4: Feature annotation problem. The tokens bracketed with broken lines are the target of syntactic annotation using UPOS and DepRel. Token-internal elements are the target of morphological annotation using FEATS (VerbForm=Inf, Polarity=Neg, Tense=Past).

syntactic functions, presumably due to a crucial lack of linguistic knowledge on the part of the annotators.

5 Toward a cross-linguistically comparable UDJ

Despite the obvious lack of linguistic adequacy as outlined above, UDJ has chosen to release its ill-formed UD treebanks at the cost of standardization and comparability. The decision is consequential: some researchers have started to raise the alarm about the danger of using UDJ for cross-linguistic comparison (Çöltekin and Rama 2022; Taguchi and Chiang 2023).

5.1 Retokenization

Taguchi and Chiang (2023) is a pioneering work that directly addresses the tokenization and feature annotation problems of UDJ. Their key insight is that the current annotation makes Japanese appear typologically similar to an isolating language like Vietnamese, rather than the agglutinative language it actually is. Recall the discussion in Section 4.2, where it was pointed out that UDJ tokenization is morpheme-based, leading to the undesirable situation in which Japanese looks like an isolating language.

To demonstrate this, Taguchi and Chiang (2023) measure morphological complexity using seven metrics from Çöltekin and Rama (2022) and show that, after retokenization, the complexity profile of Japanese shifts dramatically: the Pearson correlation with Vietnamese drops from 0.9998 to 0.67, while the correlation with Russian rises from 0.67 to 0.9992 (Table 2). The inflectional synthesis (IS) score, which is exactly 0.0 for the current treebanks—identical to Vietnamese—rises to 9.7–11.6 after retokenization, comparable to Russian (11.5) and Turkish

(13). These results provide compelling quantitative evidence that the current annotation policy fundamentally misrepresents the morphological typology of Japanese.

Table 2: Morphological complexity comparison, reproduced from [Taguchi and Chiang \(2023, 69\)](#). Pink rows: current Japanese treebanks. Green rows: retokenized versions. The IS score jumps from 0.0 to 9.7–11.6 after retokenization, shifting the typological profile from isolating to synthetic.

Morphological complexity comparison (Taguchi & Chiang, 2023, Table 3)									
Language	Typology	Treebank	TTR	MSP	WS	WH	LH	IS	MFH
Japanese	agglutinative	GSD	0.259	1.075	0.318	9.397	9.192	0.0	1.325
		GSD ₁	0.263	1.109	0.365	9.600	9.265	9.8	2.583
		GSD ₂	0.400	1.471	0.505	11.242	10.241	11.2	3.030
		GSDLUW	0.320	1.082	0.351	9.433	9.223	0.0	1.296
		GSDLUW ₁	0.338	1.061	0.448	9.600	9.455	9.7	2.619
		GSDLUW ₂	0.426	1.065	0.464	11.296	11.037	11.6	3.144
Vietnamese	analytic	VTB	0.166	1.0	0.374	9.964	9.966	0.0	1.253
English	weakly analytic	LinES, GUM	0.207	1.210	0.365	9.572	9.176	5.733	3.701
Russian	fusional	SynTagRus	0.464	1.479	0.489	11.582	10.797	11.5	3.596
Turkish	agglutinative	IMST	0.399	2.277	0.573	11.719	10.022	13	3.589

However, several major issues remain to be addressed in the work of [Taguchi and Chiang \(2023\)](#). Their method is directly affected by the drawbacks of GB, on which their merger logic is based: they take the existing SUW/LUW morpheme-level tokens produced by GB-based analysis as given, and apply rule-based rejoining to merge them back into larger tokens. The grammatical functions of the merged morphemes are then recorded as FEATS on the resulting token. Figure 7 compares the tokenization of *takakunakatta* (‘was not expensive’) in GSD, [Taguchi and Chiang \(2023\)](#), and in MUDJ.

The simple retokenization scheme of [Taguchi and Chiang \(2023\)](#) is reasonable and desirable especially if one wants to make the process fully automatic. The algorithm automatically looks up 𐀀𐀀𐀀 (using UDJ’s XP0S field) and rejoins them to the preceding token, as in:

(3) 𐀀𐀀𐀀𐀀𐀀𐀀 + 𐀀𐀀𐀀 + 𐀀𐀀 (3 tokens) → irassyara-nakat-ta (1 token)

However, the powerful retokenization algorithm leads to a serious problem such that all 𐀀𐀀𐀀 (𐀀𐀀𐀀 and 𐀀 above) are simply merged into a single word without regard to whether the preceding lexical token is a bound stem as in (3) or is a fully inflected word, as in:

(4) 𐀀𐀀𐀀 + 𐀀𐀀𐀀 + 𐀀𐀀 (3 tokens) → takaku-nakat-ta (1 token)

In the latter case, the merger is inappropriate, since the lexical token 𐀀𐀀 is a complete word on its own. Thus, the retokenized /takakunakatta/ should have been two tokens, /takaku/ + /nakatta/.² In this way, their retokenization algorithm leads to an overestimation of the morphological complexity of Japanese, as is evident from Table 2.

²A similar problem is encountered in nominal morphology as well. [Taguchi and Chiang \(2023\)](#) suggests another retokenization scheme called “level 2” where, in addition to merging verbs as noted above, case (which should be analyzed as constituting words by themselves) is also merged into a single noun word. Thus, a noun “word” becomes a case-inflected word, which is contrary to the linguistic facts of Japanese.

The simple merger approach also affects DepRel assignment, since the syntactic dependency relation between /takaku/ and /nakatta/, i.e., aux, disappears with the merger of the two words.

Although the basic idea of Taguchi and Chiang (2023)—utilizing the annotated information of the existing UDJ for retokenization and FEATS assignment—plays an important part, our Matomonized UDJ takes a novel approach to resolving the problems encountered by Taguchi and Chiang (2023). We suggest **abandoning GB and its illicit word segmentation analysis in UDJ altogether** by converting the kana-based and GB-based tokenization of UDJ into phoneme-based and linguistically adequate (hence “matomonized”) morpheme/word segmentation by means of a Python algorithm (the Matomonizer). This makes it possible to maintain the distinction between content words and functional words while merging only the internal morphemes within each word. This precise retokenization allows for preserving the dependency relation between a content word and the grammatical word depending on it.

5.2 Application to other Japonic languages

The ultimate goal of this project extends beyond Standard Japanese. Once MUDJ establishes a well-formed annotation for SJ—with correct word boundaries, UPOS, DepRel, and FEATS—the morpheme-to-word correspondence defined in MUDJ can serve as a bridge to other Japonic local languages (Figure 8). Generally speaking, Japonic languages, including SJ, share similar morphosyntactic structures, differing largely in the phonological realization of individual morphemes.

For example, as Figure 8 illustrates, SJ and Irabu Ryukyuan have similar but not identical morphosyntactic structures: *taka-ku na-kaQta* ‘was not high’ in SJ vs. *taka-f=fa njaa-t-tar* in Irabu, where the occurrence of the topic marker is obligatory and there is a dedicated negative suffix /t/.

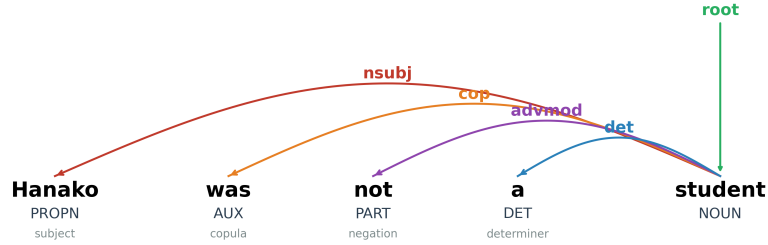
However, the two languages are similar enough for the Matomonizer to be applied. By establishing a systematic mapping between SJ and other local languages, the Matomonizer can be extended to automatically assign UPOS, DepRel, and FEATS to the data of local languages, thereby enabling the construction of UD treebanks for Japanese dialects and Ryukyuan languages—varieties that are currently unrepresented in the UD project (cf. Hirosawa 2026).

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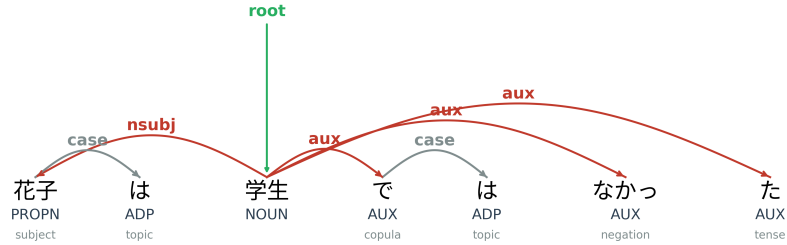
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UD English: 4 distinct DepRel labels



Current UD Japanese GSD: only aux and case



MUDJ: 4 distinct DepRel labels (cop, advmod, aux restored)

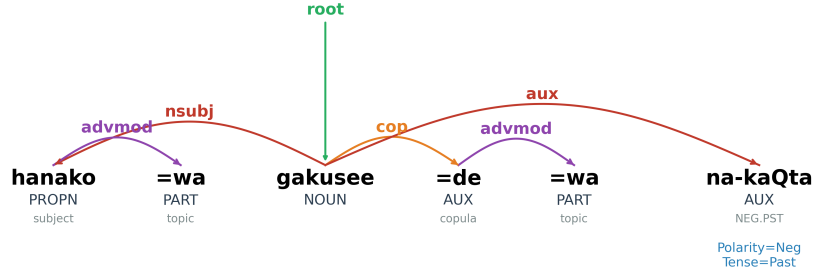


Figure 5: Dependency relation specification for *Hanako was not a student* / ~~XXXXXXXXXX~~. Top: UD English uses four distinct labels (nsubj, cop, advmod, det). Middle: the current UD Japanese GSD collapses the copula, negation, and tense into the single label aux, and both topic markers into case. Bottom: MUDJ restores the distinctions, assigning cop to the copula =de, advmod to the topic markers =wa, and aux to the negative auxiliary na-kaQta.

Taguchi & Chiang (2023): auxiliary merger logic

github.com/ctaguchi/ud_ja_standardize convert.py, lines 288–363

```
def auxiliary(self, n, sentence, i, elems):
    prev = self.table[i-1]
    if elems["XPOS"].startswith("zyodoosi"):
        try:
            feat = morphdic[elems["LEMMA"]]
        except:
            feat = []
        ...
        # KEY LINES (358-363):
        elems["COMBINED"] = True
        j = 1
        while self.table[i-j]["COMBINED"] == True:
            j += 1
        self.table[i-j]["FEATS"] += list(set(feat))
        self.table[i-j]["FORM"] += elems["FORM"]
```

* "zyodoosi" = 助動詞 (auxiliary verb in school grammar XPOS tag)

Trace for *takaku nakatta* (高くなかった):

Step	Token	UPOS	XPOS	Action	FORM result
1	takaku	ADJ	keiyoosi	not AUX: skip	takaku
2	nakatt	AUX	zyodoosi-nai	COMBINED, merge back	takakunakatt
3	ta	AUX	zyodoosi-ta	COMBINED, merge back	takakunakatta

Figure 6: The auxiliary merger logic in Taguchi and Chiang (2023)’s retokenizer (convert.py, lines 288–363, [GitHub](#)). When an AUX token with XPOS *zyodoosi* (助動詞) is encountered, it is marked as COMBINED and its FORM is appended to the nearest preceding non-combined token. Applied to *takaku nakatta*, both *nakatt* and *ta* are merged back into *takaku*, yielding a single token *takakunakatta*.

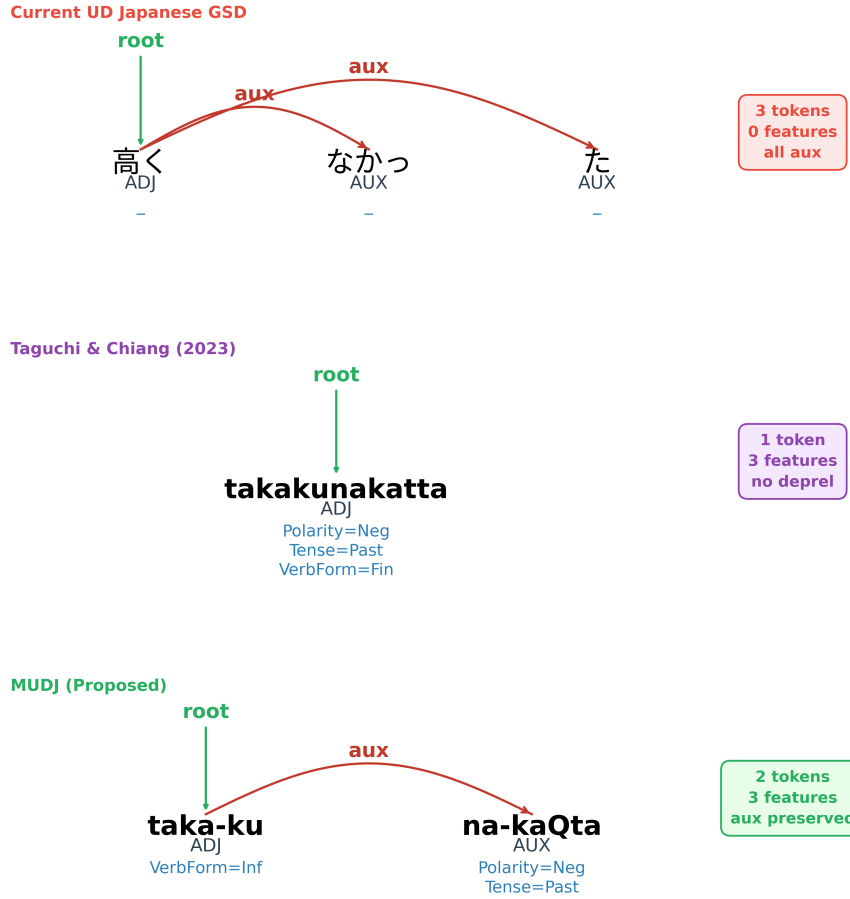


Figure 7: Taguchi and Chiang (2023) and MUDJ: a comparison. Although Taguchi and Chiang (2023) correctly analyze many instances of what should have been included as FEATS, such as Polarity=Neg and Tense=Past, their approach still encounters problems since it utilizes the existing morphological analysis in such a way that a lexical morpheme and its following grammatical morpheme(s) (as defined by UDJ) are simply merged, yielding a number of undesirable “words” like /takakunakatta/, which should have been /takaku nakatta/.

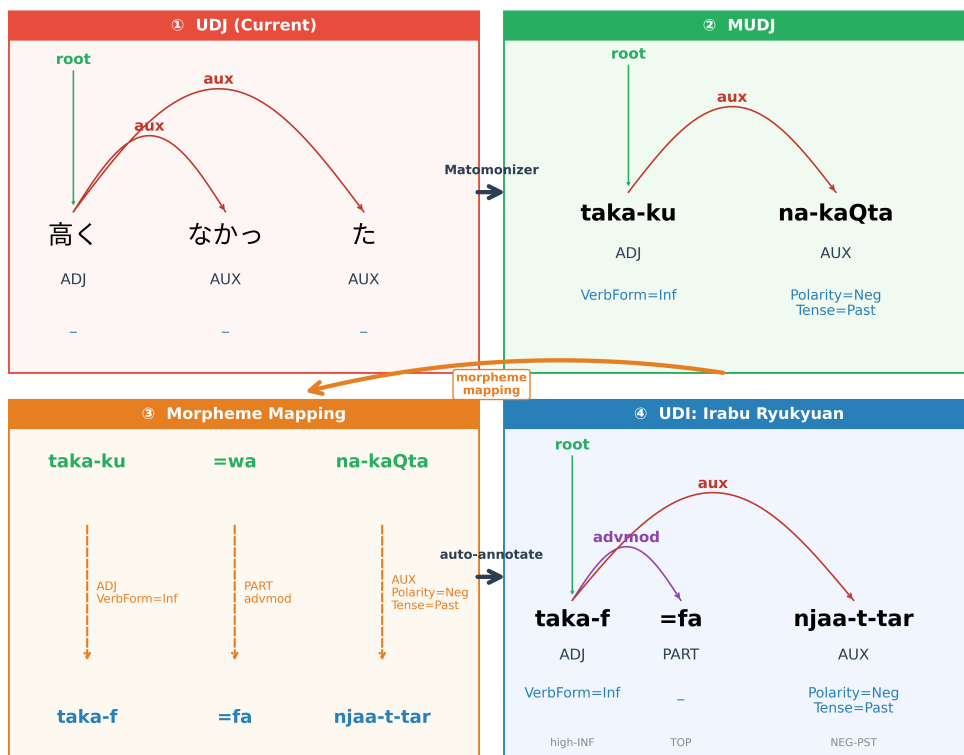


Figure 8: Project pipeline overview. (1) The current UDJ splits predicates into morpheme-level tokens with no features. (2) The Matomizer converts UDJ to MUDJ with word-level tokens and morphological features. (3) A morpheme mapping table establishes correspondences between standard and dialectal forms, transferring UPOS and DepRel. (4) The resulting UD Irabu Ryukyuan (UDI) is auto-annotated with FEATS adjusted for dialect morphology (e.g., Tense=Past → Tense=Npst; topic =wa → =fa).